CSE 4617 (SWE): AI - Quiz #4 **10 May 2024**

Duration: 30 Minutes **Full Marks: 15**

Consider that Pacman is in a world with three states X, Y, and Z. There are two possible actions from each state P and Q. Assume $\gamma = 1$, $\alpha = 0.5$, and the initial Q-values are 0 for all state.

1. We get the following samples generated from taking actions sequentially in this world: 9 (CO3) (PO3)

S	A	S'	R
X	Q	Y	2
Y	Q	Z	−6
Z	P	X	0
X	P	Y	−2
Y	P	Z	−2
Z	Q	X	2

With detailed steps, recommend the optimal policy after considering the samples.

Solution:

For $(X, Q, Y, 2)$, we get:

$$\begin{aligned} Q(X,Q) &= (1-\alpha)Q(X,Q) + \alpha\left(r + \gamma \max_A Q(Y,A)\right) \\ &= (1-0.5)\times 0 + 0.5 \times (2 + 1 \times 0) \\ &= 1 \end{aligned}$$

For $(Y, Q, Z, -6)$, we get:

$$\begin{aligned} Q(Y,Q) &= (1-\alpha)Q(Y,Q) + \alpha\left(r + \gamma \max_A Q(Z,A)\right) \\ &= (1-0.5)\times 0 + 0.5 \times (-6 + 1 \times 0) \\ &= -3 \end{aligned}$$

For $(Z, P, X, 0)$, we get:

$$\begin{aligned} Q(Z,P) &= (1-\alpha)Q(Z,P) + \alpha\left(r + \gamma \max_A Q(X,A)\right) \\ &= (1-0.5)\times 0 + 0.5 \times (0 + 1 \times 1) \\ &= 0.5 \end{aligned}$$

For $(X, P, Y, -2)$, we get:

$$\begin{aligned} Q(X,P) &= (1-\alpha)Q(X,P) + \alpha\left(r + \gamma \max_A Q(Y,A)\right) \\ &= (1-0.5)\times 0 + 0.5 \times (-2 + 1 \times 0) \\ &= -1 \end{aligned}$$

For $(Y, P, Z, -2)$, we get:

$$\begin{aligned} Q(Y,P) &= (1-\alpha)Q(Y,P) + \alpha\left(r + \gamma \max_A Q(Z,A)\right) \\ &= (1-0.5)\times 0 + 0.5 \times (-2 + 1 \times 0.5) \\ &= -0.75 \end{aligned}$$

For $(Z, Q, X, 2)$, we get:

$$\begin{aligned} Q(Z,Q) &= (1-\alpha)Q(Z,Q) + \alpha\left(r + \gamma \max_A Q(X,A)\right) \\ &= (1-0.5)\times 0 + 0.5 \times (2 + 1 \times 1) \\ &= 1.5 \end{aligned}$$

So the optimal actions for each state are:

$$\begin{aligned} \pi(X) &= \operatorname{argmax}\,(Q(X,Q), Q(X,P)) = Q \\ \pi(Y) &= \operatorname{argmax}\,(Q(Y,Q), Q(Y,P)) = P \\ \pi(Z) &= \operatorname{argmax}\,(Q(Z,Q), Q(Z,P)) = Q \end{aligned}$$

Rubric:
- 1 point per iteration
- 1 point per optimal action

2. Assume that after the previous steps, we decide to utilize two features: 6 (CO3) (PO3)

$$f_1(s,a) = 1$$

$$f_2(s,a) = \begin{cases} 1 & a = P \\ -1 & a = Q \end{cases}$$

to calculate the value of a state and the initial weights $w_1 = w_2 = 1$. The initial Q-values are the once determined in Question 1. We observe the following samples:

S	A	S'	R
X	P	Y	4
Y	Q	X	0

With detailed steps, recommend the optimal policy after considering the samples.

Solution:

For $(X, P, Y, 4)$, we get:

$$\begin{aligned}
\hat{Q}(X,P) &= w_1 f_1(X,P) + w_2 f_2(X,P) \\
&= 1 \times 1 + 1 \times 1 \\
&= 2 \\
\text{diff} &= \left[r + \max_A Q(Y,A)\right] - \hat{Q}(X,P) \\
&= 4 - 0.75 - 2 \\
&= 1.25 \\
w_1 &= w_1 + \alpha[\text{diff}] f_1 \\
&= 1 + 0.5 \times 1.25 \times 1 \\
&= 1.625 \\
w_2 &= w_2 + \alpha[\text{diff}] f_2 \\
&= 1 + 0.5 \times 1.25 \times 1 \\
&= 1.625
\end{aligned}$$

For $(Y, Q, X, 0)$, we get:

$$\begin{aligned}
\hat{Q}(Y,Q) &= 1.625 \times 1 + 1.625 \times (-1) \\
&= 0 \\
\text{diff} &= \left[r + \max_A Q(X,A)\right] - \hat{Q}(Y,Q) \\
&= 0 + 1 - 0 \\
&= 1 \\
w_1 &= w_1 + \alpha[\text{diff}] f_1 \\
&= 1.625 + 0.5 \times 1 \times 1 \\
&= 2.125 \\
w_2 &= w_2 + \alpha[\text{diff}] f_2 \\
&= 1.625 + 0.5 \times 1 \times (-1) \\
&= 1.125
\end{aligned}$$

The Q-values are:

$$\begin{aligned}
\hat{Q}(X,P) &= w_1 f_1(X,P) + w_2 f_2(X,P) \\
&= 2.125 \times 1 + 1.125 \times 1 \\
&= 3.25 \\
\hat{Q}(X,Q) &= w_1 f_1(X,P) + w_2 f_2(X,Q) \\
&= 2.125 \times 1 + 1.125 \times (-1) \\
&= 1
\end{aligned}$$

Note that, the features do not depend on the states, so the Q-values will be the same for all three. We get:

$$\pi(X) = \pi(Y) = \pi(Z) = \text{argmax}(Q(X,P), Q(X,Q)) = P$$

Rubric:

- 1 point per update
- 1 point for Q-values
- 1 point per optimal policy